URBAN
GLOW
GRID

GENAI ASSISTANT DOSSIER

How We Worked with AI

Documenting our collaborative innovation process for the Urban Glow Grid project.

Abir ISLAM
CEO / Simulation

Sofiane LACHHAB
CTO / Planning

Mohamed MELLOUK
Creative Director

1. Prompt & Template Library

Our library consists of 10 specialized prompts and 2 reusable templates designed for the Urban Glow Grid project.

USE CASE 1: RESEARCH & DATA

PROMPT #1: ENERGY METRICS

```
"Find the average peak-hour energy consumption for the 8th arrondissement
of Paris. Provide the data in a simplified ratio for a game simulation."
```

PROMPT #2: CO2 CALIBRATION

```
"What is the average CO2 emission per kWh in the French grid? We need this
for a 'Sustainability' feature in our dashboard."
```

USE CASE 2: IDEATION & SPATIAL PLANNING

PROMPT #3: PHYSICAL MODEL SKETCH

```
"We are building a wooden scale model. Suggest a way to represent 'energy
glow' using fiber optics. How should we carve the wood?"
```

PROMPT #4: IOT SENSORS

```
"Which IoT sensors are best for tracking energy data in a scale model?
Suggest positioning for optimal visibility."
```

USE CASE 3: PLANNING & VALIDATION

PROMPT #5: SPRINT PLANNING

```
"Draft a 4-week sprint for a hybrid project (Physical Model + Simulation).
Include a 'Convergence Week' for syncing."
```

PROMPT #6: LOGIC VALIDATION

`"Review this JS logic: if totalLoad > capacity * 1.2, trigger a blackout. Is this too punishing? Suggest a 'warning' state."`

USE CASE 4: CODING & TECHNICAL DESIGN

PROMPT #7: SVG OPTIMIZATION

`"Optimize this SVG path of the Seine river for a web simulator. The current path is too complex and causes lag."`

PROMPT #8: ACCESSIBILITY

`"How can we make our game accessible to color-blind users? Suggest a palette for the energy bar."`

USE CASE 5: CONTENT & CRITICISM

PROMPT #9: PITCH SCRIPT

`"Write a 1-minute pitch script for 'Urban Glow Grid'. Focus on the contrast between dark city and glowing districts."`

PROMPT #10: JURY REVIEW

`"Act as a critical jury member. Identify logic gaps in our value proposition for Urban Glow Grid."`

REUSABLE TEMPLATES

TEMPLATE A: LYRA (MASTER OPTIMIZER)

Role: Lyra, Expert in Prompt Engineering. **Method:** 4-D (Deconstruct, Diagnose, Develop, Deliver).
Used to transform vague ideas into high-precision technical blueprints.

TEMPLATE B: THE DIRECTOR'S MINDSET

Role: Technical Systems Director. **Mission:** Design the bridge between Digital and Physical layers.

2. Conversation Excerpts

Excerpt #1 — LED Color Logic & Breathing Effect (Abir · Gemini)

Je développe un simulateur web appelé 'Urban Glow Grid' pour visualiser la consommation d'énergie de Paris. Écris-moi une fonction JavaScript qui fait varier la couleur de remplissage des bâtiments du vert au rouge pulsé...

Afficher le raisonnement

C'est une excellente idée d'ajouter cette dimension organique à **Urban Glow-Grid**. Visualiser les données de manière viscérale crée une urgence émotionnelle que de simples graphiques ne peuvent pas transmettre.

Pour obtenir cet effet subtil de "respiration", la meilleure approche est de combiner JavaScript pour la logique d'état et l'interpolation des couleurs, avec les variables CSS pour gérer l'animation matériellement accélérée.

✓ **What worked:** Gemini proposed combining JS for state logic with CSS Variables for hardware-accelerated color interpolation. The "breathing" metaphor was directly implemented in `Urban_GlowGrid_Simulator.html`.

✗ **Human Override:** The AI initially suggested a complex mathematical approach for the oscillation. We simplified this to a clean CSS transition to ensure smooth performance across all browsers.

Excerpt #2 — Carbon Intensity Data (Abir · Gemini)

Quelle est l'émission moyenne de CO2 par kWh pour le réseau électrique français en 2023 ? Comment ce chiffre évolue-t-il pendant les heures de pointe...

C'est une excellente initiative pour ton projet étudiant. Prendre en compte la variation temporelle de l'intensité carbone est la marque d'un simulateur de haute précision. La plupart des bilans se contentent d'une moyenne annuelle, ce qui masque l'impact réel des pics de demande urbaine.

Voici les chiffres réels et précis pour la France, basés sur les données du **Réseau de Transport d'Électricité (RTE)** et de l'**ADEME**.

✓ **Data Source:** The AI correctly referenced **RTE** and **ADEME**. We used the official figure of ~52g CO_2/kWh (average for France) to calibrate our sustainability scores.

Lesson: AI is excellent for identifying correct sources (RTE/ADEME), but figures were cross-checked manually against official PDF reports to ensure accuracy.

Excerpt #3 — Scale & Physical Constraints (Sofiane · Gemini)

Je souhaite construire une maquette physique en bois des arrondissements de Paris sur une planche qui fait exactement 80×60cm. Paris fait environ 10km de large. Quelle échelle dois-je utiliser...

C'est une excellente question technique pour ton projet **Urban Glow-Grid**. Le choix de l'échelle est le compromis le plus critique entre la lisibilité de la maquette et la complexité de la fabrication.

1. Analyse des dimensions et calcul de l'échelle

Paris intra-muros mesure environ **12 km d'Est en Ouest** et environ **9 km du Nord au Sud**.

✗ **Constraint Issue:** The AI provided an accurate analysis of Paris dimensions (12km x 9km) but the initial scale suggestion was too complex for a 3-week timeline.

✓ **Correction:** We manually simplified the scale to 1:15,000 to fit our 80×60cm board, prioritizing readability for the jury. Recorded in Weekly_Meeting_2.pdf .

3. Experiment Log

Experiment	Tool	Iterations	Outcome	Linked Artifact
JS Simulation Core	ChatGPT	4	Accepted	Simulator.html
Physical Map Sketch	Claude	3	Accepted (simplified)	Meeting_2.pdf – p.3
Pitch Slogan	Gemini	10+	Rejected — human-written	Pitch – Slide 1
PDF Gen Script	Antigravity	5	Accepted	generate_dossier_pdf.py
Game Mechanic Ideation	ChatGPT	1 diverge + 2 converge	Accepted — "Glow" selected	Pitch – Slide 6

3.1 — DIVERGE & CONVERGE: SELECTING THE "GLOW" MECHANIC

Donne-moi 10 mécaniques de jeu distinctes et créatives qui rendent l'économie d'énergie visuellement gratifiante...

Pour **Urban Glow-Grid**, l'enjeu est de transformer une contrainte (la sobriété) en une force vitale. Voici 10 mécaniques :

1. La "Respiration Lumineuse" (Bio-Luminescence)
Au lieu d'un éclairage statique, les zones sobres adoptent un effet de "pulsation lente"...

↗ DIVERGE — 10 MECHANICS GENERATED

Parmi ces 10 idées, concentrons-nous sur 'Glow Intensity'. Comment cela fonctionnerait-il concrètement ?

C'est une excellente question de design comportemental. En psychologie cognitive, on sait que la **récompense variable** et le **renforcement positif** sont bien plus efficaces pour l'apprentissage à long terme que la punition.

↘ CONVERGE — 1 MECHANIC RETAINED

Why "Glow Intensity" won: Positive reinforcement and variable reward systems are more effective for learning than punishment. Implemented in Pitch Slides – Slide 6 .

4. Project Integration

AI was used at three levels: the **digital simulator**, the **physical model**, and the **narrative**. Each contribution below is linked to a concrete project artifact.

4.1 — AI CONTRIBUTIONS LINKED TO PROJECT ARTIFACTS

Member	AI Contribution	Linked Artifact	Override
Abir ISLAM	Multi-state energy logic (Green→Blackout), CSS pulse animation, PDF automation	Simulator.html – L.89–134 generate_dossier_pdf.py	~30%
Sofiane LACHHAB	Wood-cutting zone layout (5 zones), LED groove depth, fiber-optic routing	Weekly_Meeting_2.pdf – p.3 Pitch Slides – Slide 8	~50%
Mohamed MELLOUK	"Glow" mechanic ideation, pitch narrative reframe ("mindfulness"), visual palette	Pitch Slides – Slides 4–6 Simulator.html – UI	~60%

⚠ **Key Takeaways**

- **Best use:** First-draft generation, technical scaffolding, and documentation automation.
- **Limits:** AI over-engineered solutions and ignored explicit constraints — heavy filtering was always required.
- **Rule:** Every AI output was verified against a real source or tested in code before integration.

5. Glossary of AI Terms

Prompt Engineering — Crafting input text deliberately to guide an AI toward a specific, quality output using role, format, and constraints.

Few-Shot Prompting — Providing 2–5 examples inside the prompt so the model understands the expected format or style.

Chain-of-Thought (CoT) — Asking the AI to reason step-by-step before giving a final answer, improving accuracy on multi-step problems.

Hallucination — When an AI confidently generates false information. All AI-produced data was cross-checked against official sources.

Context Window — The maximum amount of text an AI can process at once. In long sessions, earlier constraints can be forgotten.

Human-in-the-Loop (HITL) — Every AI output was reviewed and validated by a team member before being used in the project.

Iteration / Refinement Loop — The process of repeatedly improving AI output through feedback. Most usable results required 3–5+ rounds.

6. Individual Reflections

Abir ISLAM

CEO

I used assistants to generate the simulation logic in JS to test how our physical LEDs should react to different energy loads.

✓ SUCCESS

A major success was the automation of our file generation.

✗ FAILURE

A failure was when the AI suggested overly complex math.

→ LESSON LEARNED

I learned that AI is a powerful accelerator for technical prototyping but requires a strong "filter".

Sofiane LACHHAB

CTO

I used AI as a spatial planning assistant to sketch the wood-cutting map of Paris for our physical installation.

✓ SUCCESS

A success was the advice on how to integrate LED wiring within the wooden structure.

✗ FAILURE

A failure was when it suggested a 1:1 scale that was physically impossible.

→ LESSON LEARNED

I learned that AI is excellent for architectural brainstorming.

Mohamed MELLOUK

CREATIVE DIRECTOR

My role was to find game mechanics that encourage energy mindfulness and to select the specific visual assets for our UI.

✓ SUCCESS

A success was the "Glow" mechanic, which makes energy saving visually rewarding.

→ LESSON LEARNED

I learned that AI is a great ideation partner for game systems, but the final "vibe" must remain human-led.